

Husayn El Sharif

Senior Data Scientist • Machine Learning Engineer

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Professional Summary

Data Scientist and ML Engineer with 10+ years of experience building data-driven solutions across environmental, engineering, and business applications. Combines a Ph.D. in Civil Engineering from Georgia Tech with hands-on expertise in machine learning, Generative AI, Retrieval-Augmented Generation (RAG), deep learning, geospatial analytics, and cloud-based data platforms. Google Cloud Professional Machine Learning Engineer and Databricks Certified Generative AI Engineer Associate.

Technical Skills

Programming	Python, SQL, R, MATLAB
Machine Learning	Scikit-Learn, TensorFlow, PyTorch, XGBoost, Deep Learning
Data Engineering & MLOps	Numpy, Pandas, PySpark, MLflow, Docker, Airflow, Streamlit, Git
AI Engineering	Prompt Engineering, Fine-Tuning, Agentic RAG, Claude Code, Codex
Cloud & Data Platforms	Google Cloud, AWS, Databricks, Snowflake, Hugging Face
Visualization	Matplotlib, Seaborn, Plotly, Tableau, Power BI
Geospatial Analysis	ArcGIS, QGIS, GeoPandas, Google Earth Engine

Certifications

Google Cloud Professional Machine Learning Engineer (2026)

Databricks Certified Generative AI Engineer Associate (2026)

Selected Machine Learning Projects

Remote Sensing & ML for Water Quality Assessment at Lake Lanier

- Built a satellite-based ML pipeline with Sentinel-2 and Google Earth Engine to estimate chlorophyll-a concentrations and detect harmful algal blooms, enabling near-real-time water quality monitoring for Gwinnett County Department of Water Resources. [View Project](#)

Smart Meter Risk Segmentation with Unsupervised ML (AMI Data)

- Built a k-means clustering pipeline on high-frequency AMI data to segment smart meters by load and power-quality behavior, enabling identification of high-risk assets and supporting proactive maintenance prioritization for utility operations. [View Project](#)

AI-Powered Retinal Disease Detection with Deep Learning & Computer Vision Pipeline

- Developed a deep learning computer vision pipeline for retinal disease classification, achieving 89% accuracy and 0.88 macro-F1 score across four diagnostic classes, supporting automated screening and clinical decision support. [View Project](#)

Agentic RAG Workflow for Building & Fire Code Review

- Built a LangChain agentic RAG app with hybrid keyword-semantic search for Cobb County building/fire code Q&A across ordinances, permit forms, and code references; improved faithfulness from 0.26 to 0.97, achieving 0.77 context precision and 0.72 context recall. [View Project](#)

Professional Experience

Senior Data Scientist (Part-Time)
Demand Delta

Marietta, GA | Sep 2025 – Present

- Developed a property lead-scoring pipeline using weather, permit, valuation, and geospatial data; outperformed a heuristic baseline with logistic regression/XGBoost, identifying a projected 12% revenue lift (~\$450K per 600 leads) and 3.3x higher PR-AUC. Added Platt scaling and SHAP for explainable, per-lead insights.
- Built an explainable RAG platform to match project requirements with qualified engineering firms, indexing 7K+ public profiles into ~70K retrieval documents using PostgreSQL/pgvector, embeddings, hybrid search, RRF, and cross-encoder reranking.
- Built an AI-powered lead enrichment pipeline (Python, Google Gemini, Google Places API) to surface high-value industrial prospects for an engineering consultancy, replacing manual prospecting with a scored outreach list and accelerating lead qualification.

Senior Data Scientist (Research Engineer)
Georgia Institute of Technology, School of Civil & Environmental Engineering
Atlanta, GA | Aug 2012 – Present

- Built Dockerized environments for geospatial ML workflows supporting reproducible deployment of satellite-based environmental monitoring pipelines.
- Developed ML models integrating soil, crop, and climate data to predict crop yield and irrigation demand, enabling data-driven water resource planning and quantifying long-term risks to agricultural production in the Apalachicola-Chattahoochee-Flint River Basin.
- Oversaw \$400K+ in annual federal and state research funding as Assistant Director of the Georgia Water Resources Institute.
- Mentored graduate researchers in data science and ML methods, supporting publications and conference presentations.

Engineering Consultant (Part-Time)
4Earth, Inc.

Kennesaw, GA | Aug 2022 – Aug 2025

- Developed physics and data-driven system models and AI-Ops pipelines for real-time AWS IoT sensor analytics, improving engineering efficiency and reducing time-to-market by 20%.

Data Scientist (Graduate Research Assistant)
Florida Atlantic University, Department of Civil, Environmental, & Geomatics Engineering
Boca Raton, FL | May 2010 – Aug 2012

- Built and validated ML models to reconstruct missing environmental data, improving accuracy and spatial consistency for hydrologic and climate analyses.

Education

Ph.D. Civil Engineering	Georgia Institute of Technology	Atlanta, GA	GPA: 3.97	May 2019
M.Sc. Civil Engineering	Georgia Institute of Technology	Atlanta, GA	GPA: 3.97	May 2016
B.Sc. Civil Engineering	Florida Atlantic University	Boca Raton, FL	GPA: 3.81	Aug 2011